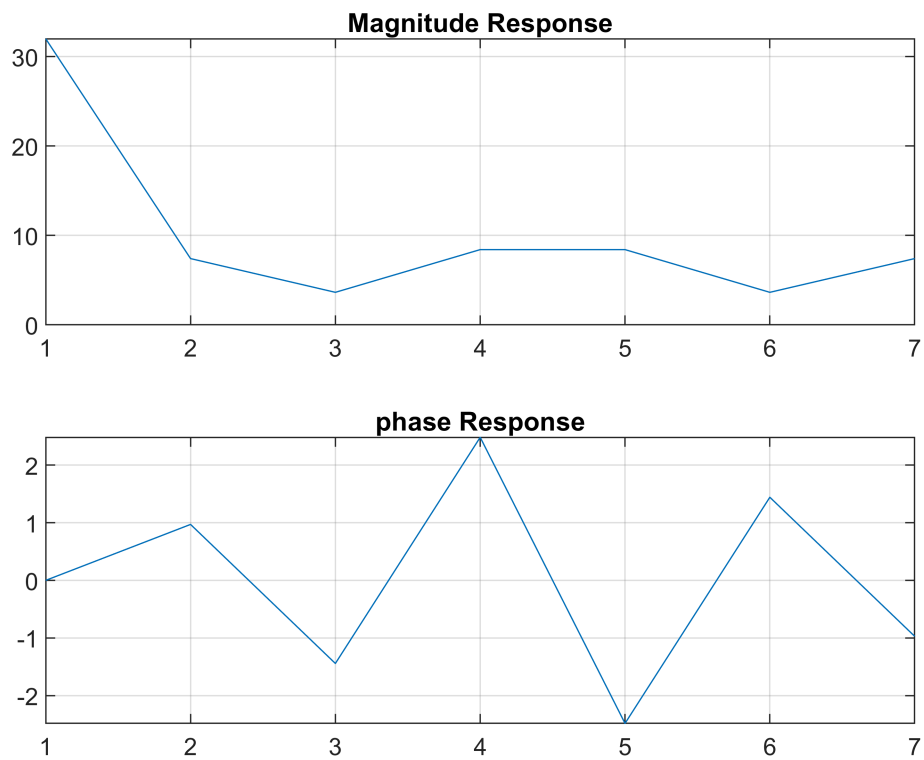
```
%USING INBUILT FUNCTION
n = 0:6;
x = [4 6 2 1 7 4 8]; % function is given
a = fft(x);
mag = abs(a);
pha = angle(a);
subplot(2,1,1);
plot(mag);
grid on
title('Magnitude Response');
subplot(2,1,2);
plot(pha);
grid on
title('phase Response');
```




```
%N-POINT FFT WITHOUT USING INBUILT FUNCTION OF USER DEFINED FUNCTION
x=input('Enter the sequence x[n]= ');
N=input('Enter the value N point= ');
L=length(x);
x_n=[x,zeros(1,N-L)];
for i=1:N
for j=1:N
temp=-2*pi*(i-1)*(j-1)/N;
DFT_mat(i,j)=exp(complex(0,temp));
end
```

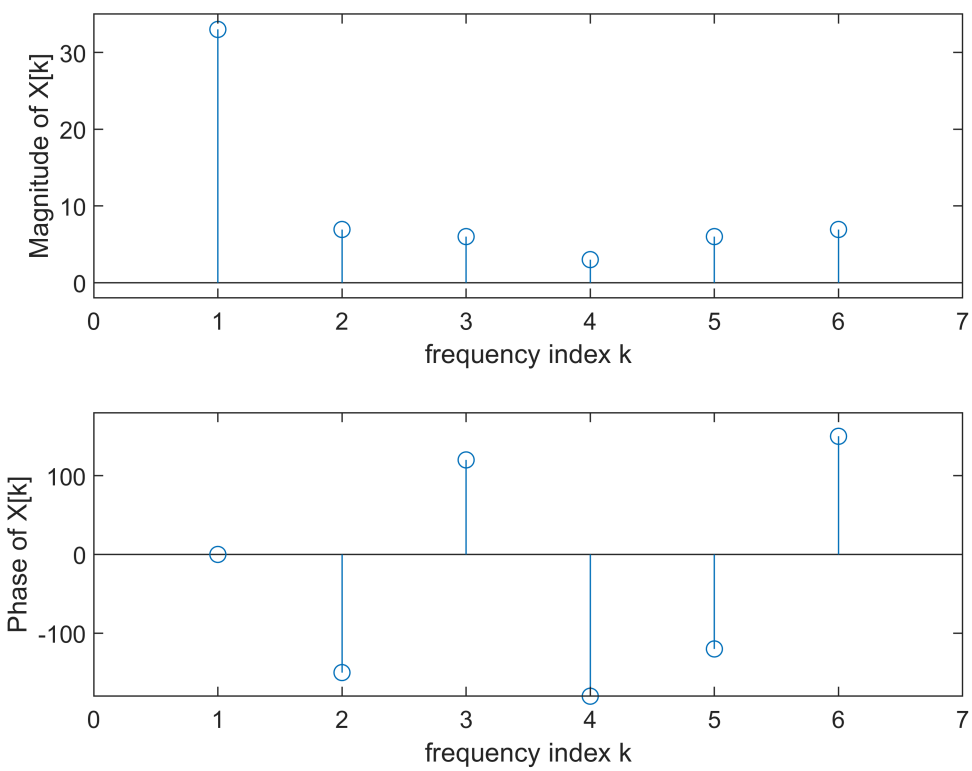
```
end
X_k=DFT_mat*x_n';
disp('N point DFT is X[k] = ');
```

```
N point DFT is X[k] =
```

```
disp(X_k);
```

```
   33.0000 + 0.0000i
   -6.0000 - 3.4641i
   -3.0000 + 5.1962i
   -3.0000 - 0.0000i
   -3.0000 - 5.1962i
   -6.0000 + 3.4641i
```

```
mag=abs(X_k);
phase=angle(X_k)*180/pi;
subplot(2,1,1);
stem(mag);
xlabel('frequency index k');
ylabel('Magnitude of X[k]');
axis([0 N+1 -2 max(mag)+2]);
subplot(2,1,2);
stem(phase);
xlabel('frequency index k');
ylabel('Phase of X[k]');
axis([0 N+1 -180 180]);
```



```
%For user defined sequence :
x=input('Enter the sequence :-');
n=input('Enter the length of fft :-'); %compute fft
disp('fourier transformed signal');
```

```
fourier transformed signal
```

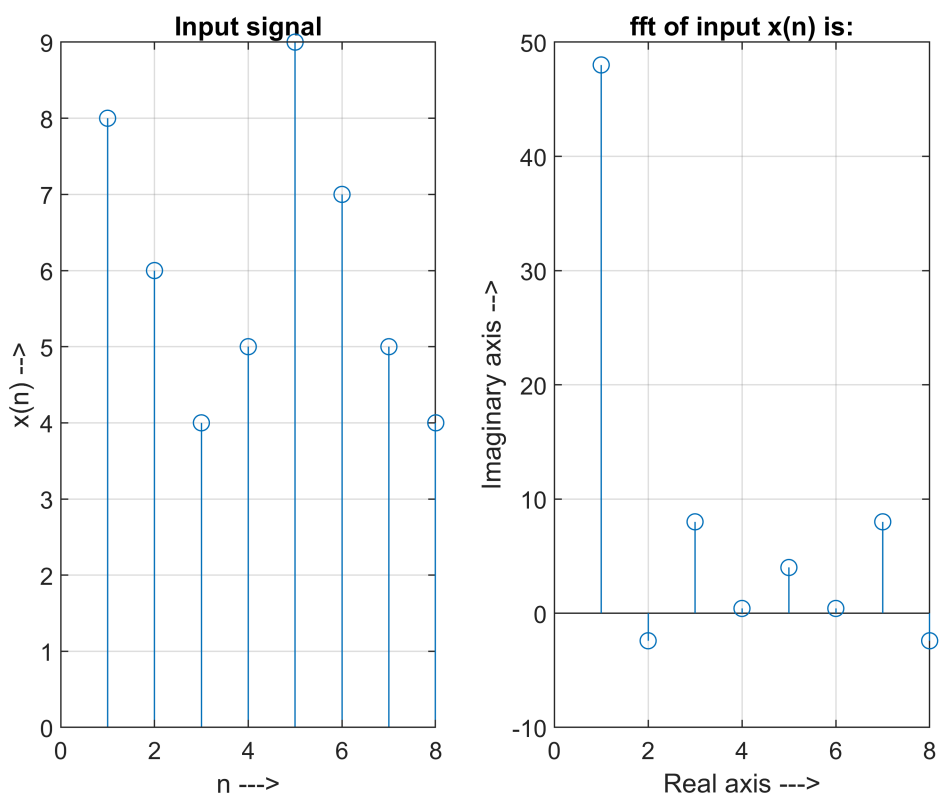
```
X=fft(x,n)
```

```
X = 1×8 complex
  48.0000 + 0.0000i  -2.4142 + 1.0000i   8.0000 - 4.0000i   0.4142 - 1.0000i ···
```

```
subplot(1,2,1);stem(x);
title('Input signal');
xlabel('n --->');
ylabel('x(n) -->');grid;
subplot(1,2,2);stem(X);
```

```
Warning: Using only the real component of complex data.
```

```
title('fft of input x(n) is:');
xlabel('Real axis --->');
ylabel('Imaginary axis -->');
grid;
```



```
%For user defined sequence :
x=input('Enter the sequence :-');
n=input('Enter the length of fft :-'); %compute fft
disp('Fourier transformed signal');
```

```
Fourier transformed signal
```

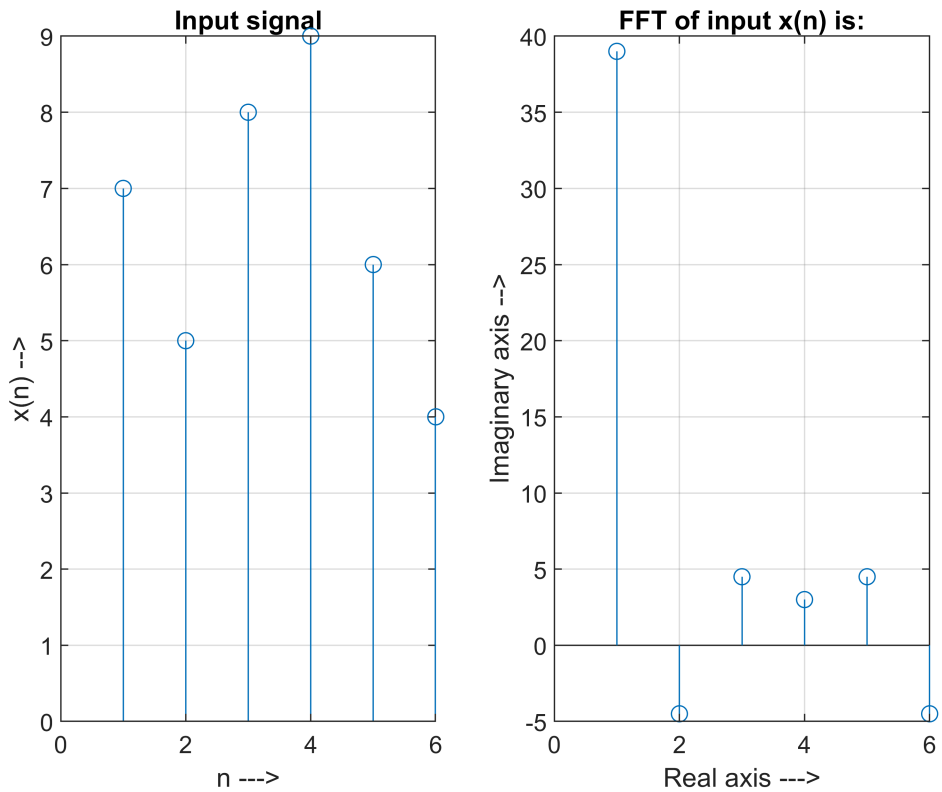
```
X=fft(x,n)
```

```
X = 1×6 complex
  39.0000 + 0.0000i  -4.5000 - 2.5981i   4.5000 + 0.8660i   3.0000 + 0.0000i ···
```

```
subplot(1,2,1);
stem(x);
title('Input signal');
xlabel('n --->');
ylabel('x(n) -->');
grid;
subplot(1,2,2);stem(X);
```

```
Warning: Using only the real component of complex data.
```

```
title('FFT of input x(n) is:');
xlabel('Real axis --->');
ylabel('Imaginary axis -->');
grid;
```




```
%For user defined sequence :
x=input('Enter the sequence :-');
n=input('Enter the length of fft :-'); %compute fft
disp('Fourier transformed signal');
```

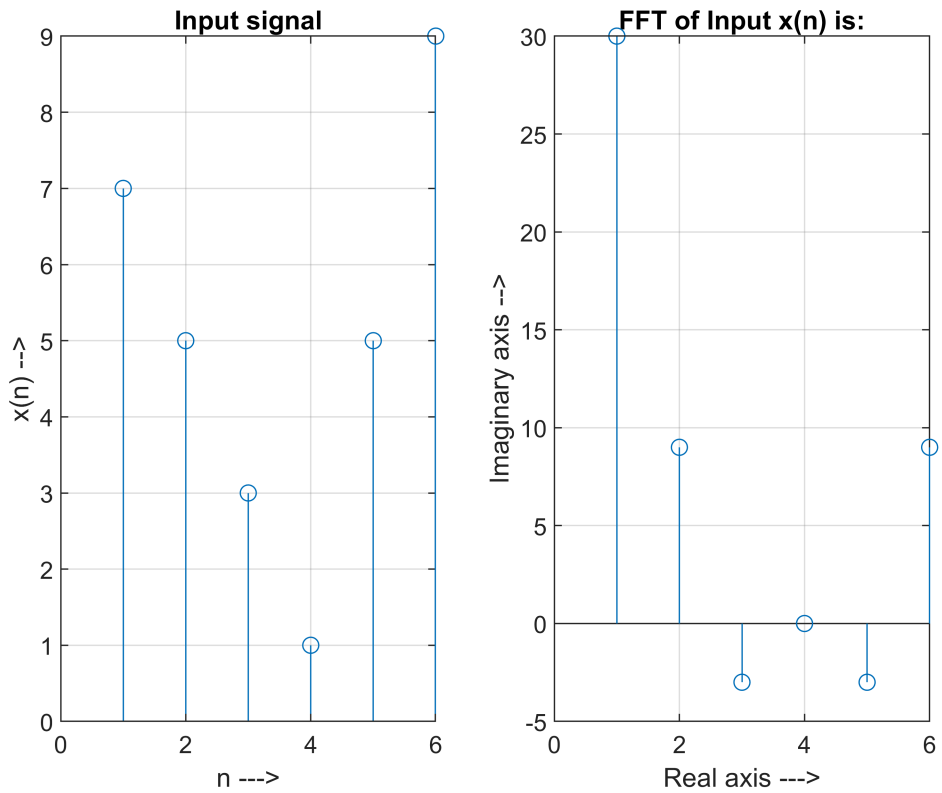
```
Fourier transformed signal
```

```
X=fft(x,n)
```

```
X = 1×6 complex
  30.0000 + 0.0000i   9.0000 + 5.1962i  -3.0000 + 1.7321i   0.0000 + 0.0000i ···
```

```
subplot(1,2,1);
stem(x);
title('Input signal');
xlabel('n --->');
ylabel('x(n) -->');
grid;
subplot(1,2,2);
stem(X);
```

```
Warning: Using only the real component of complex data.
```

```
title('FFT of Input x(n) is:');
xlabel('Real axis --->');
ylabel('Imaginary axis -->');
```

```
grid;
```